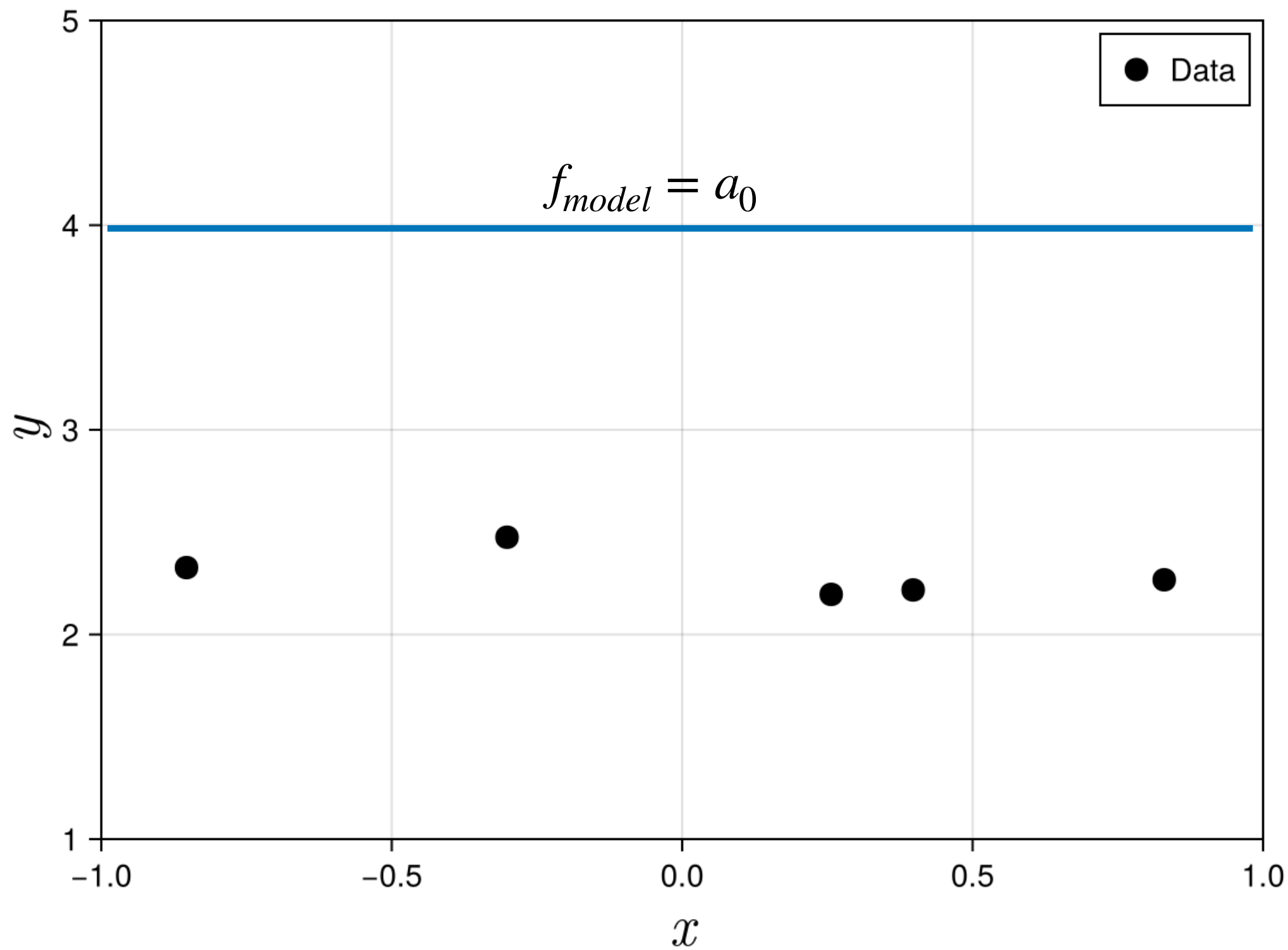
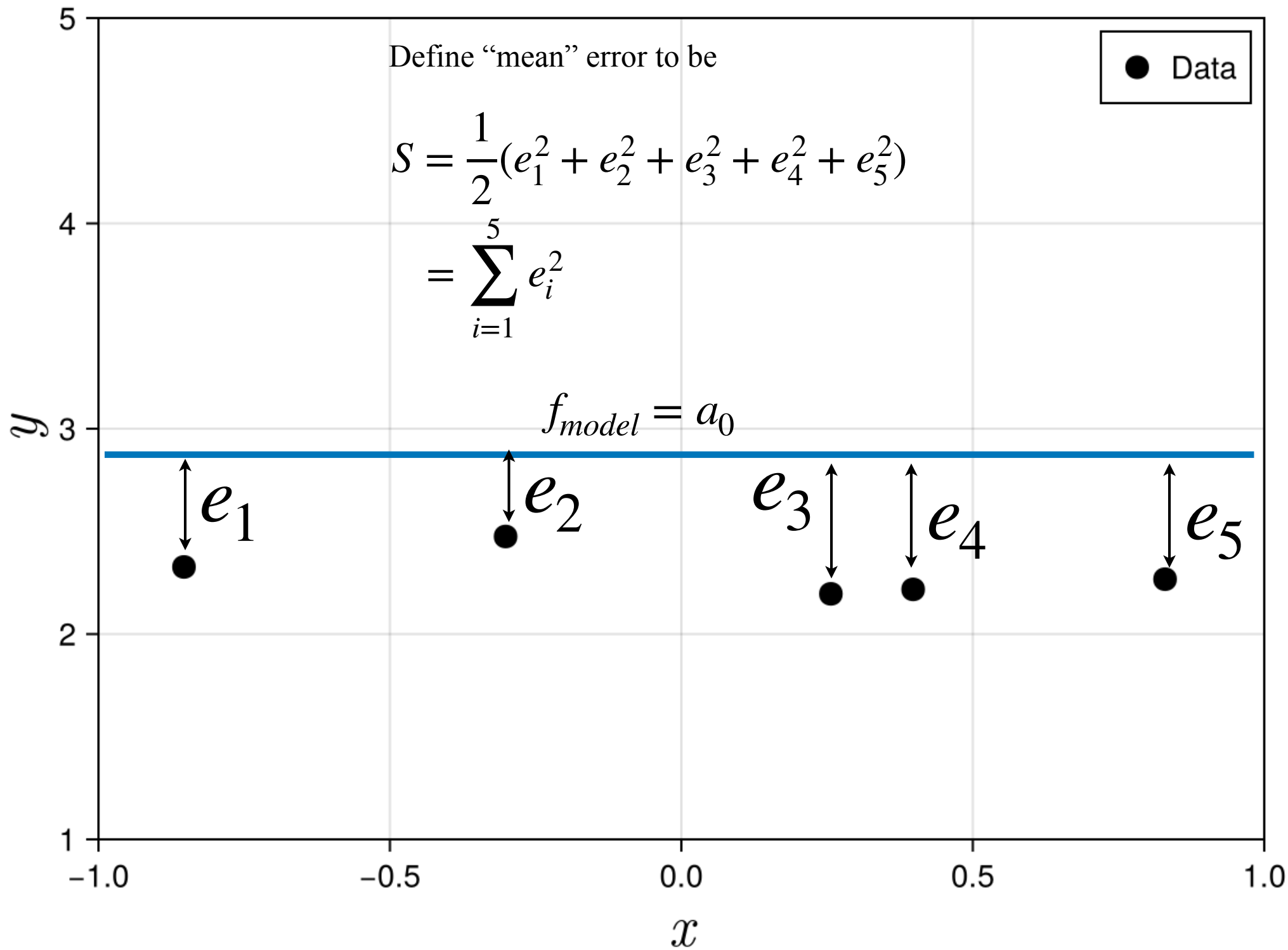


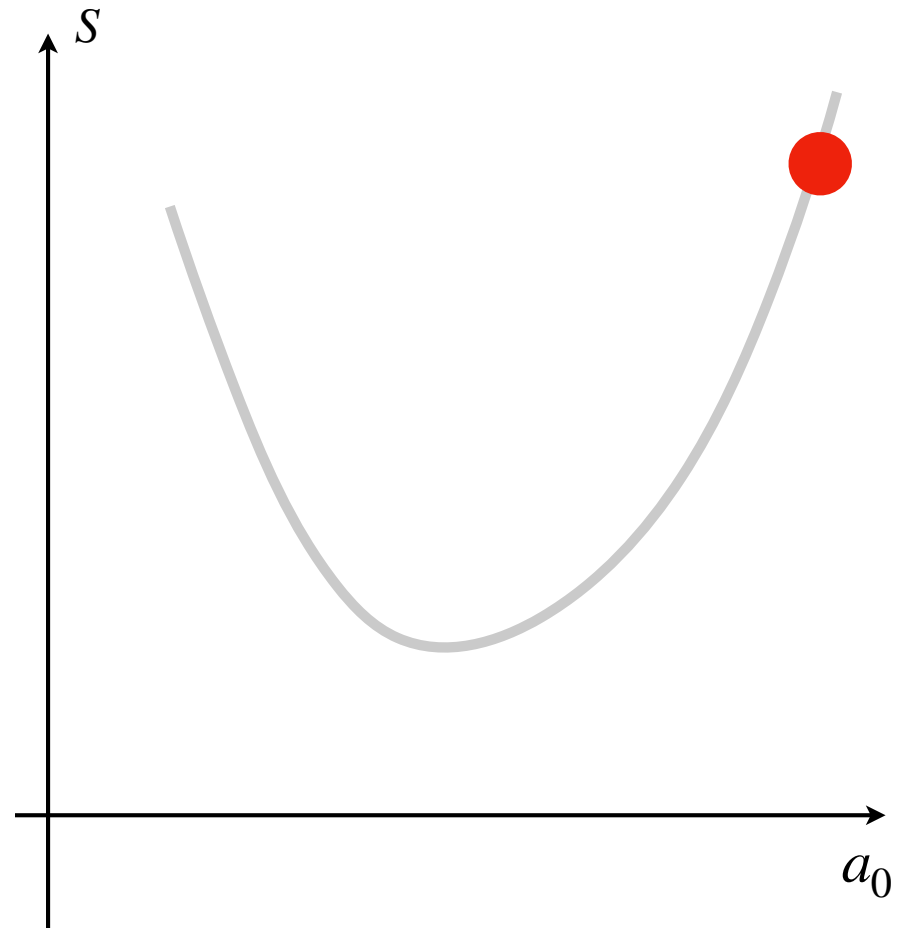
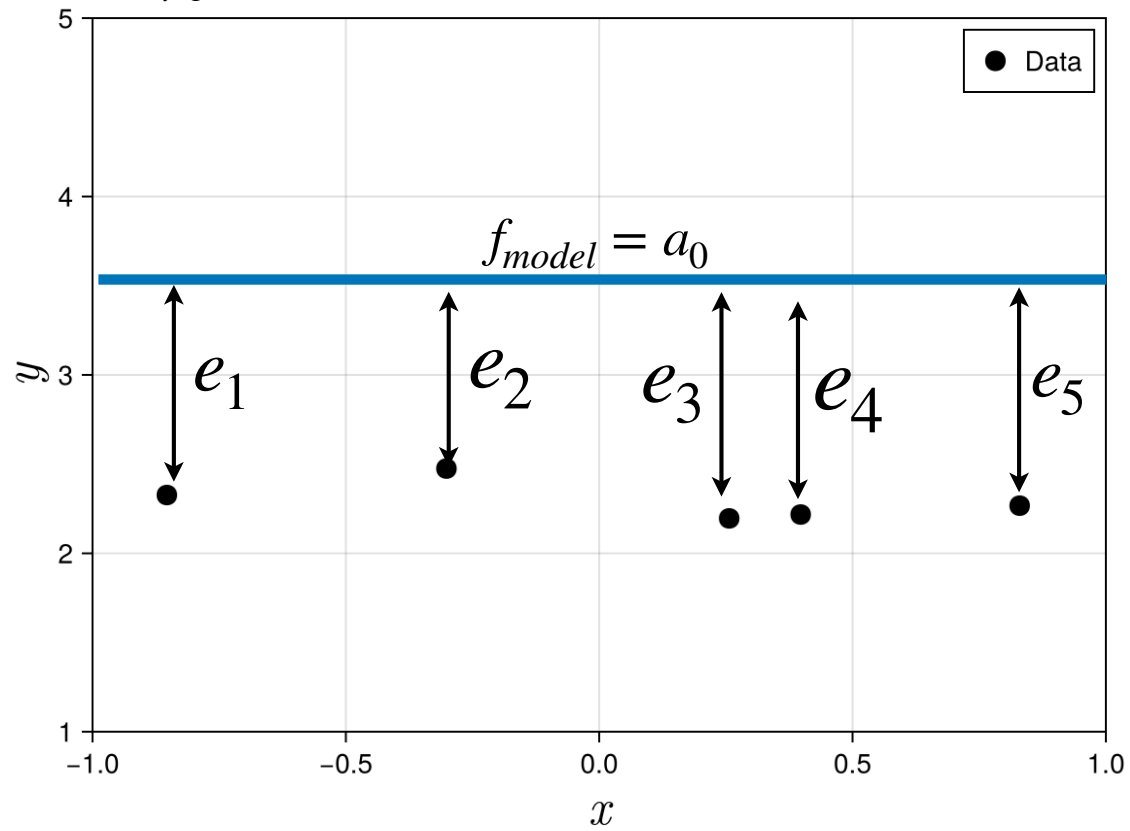
The Idea!





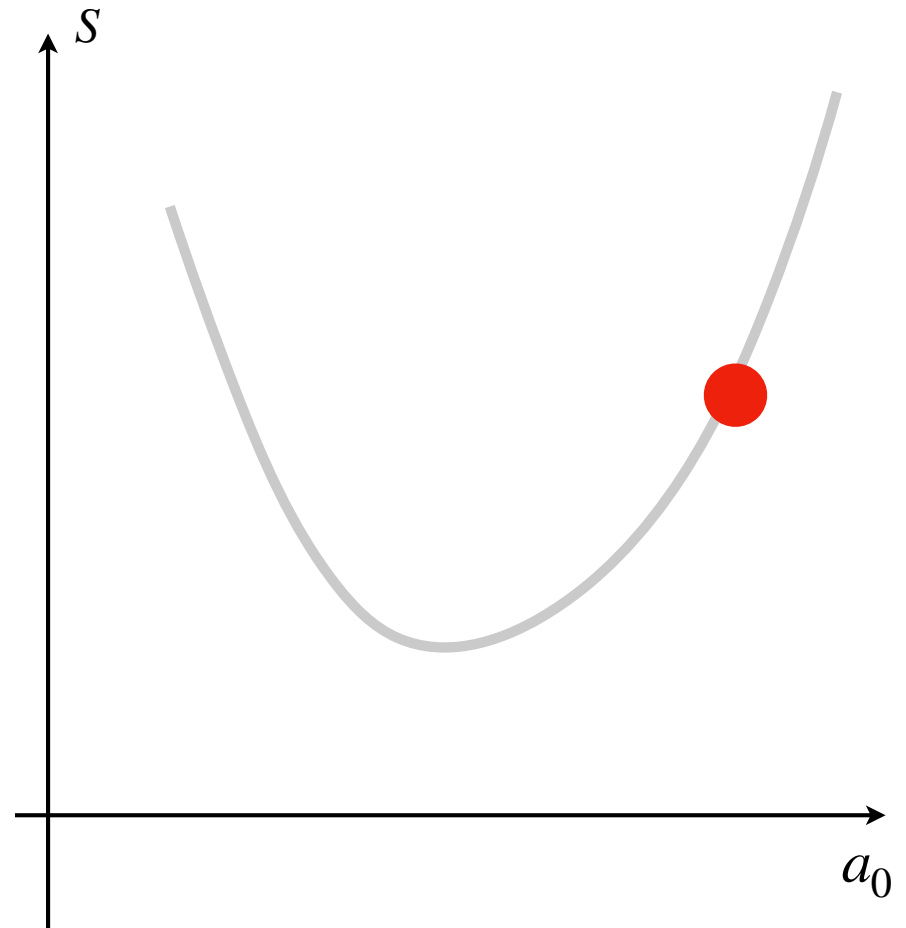
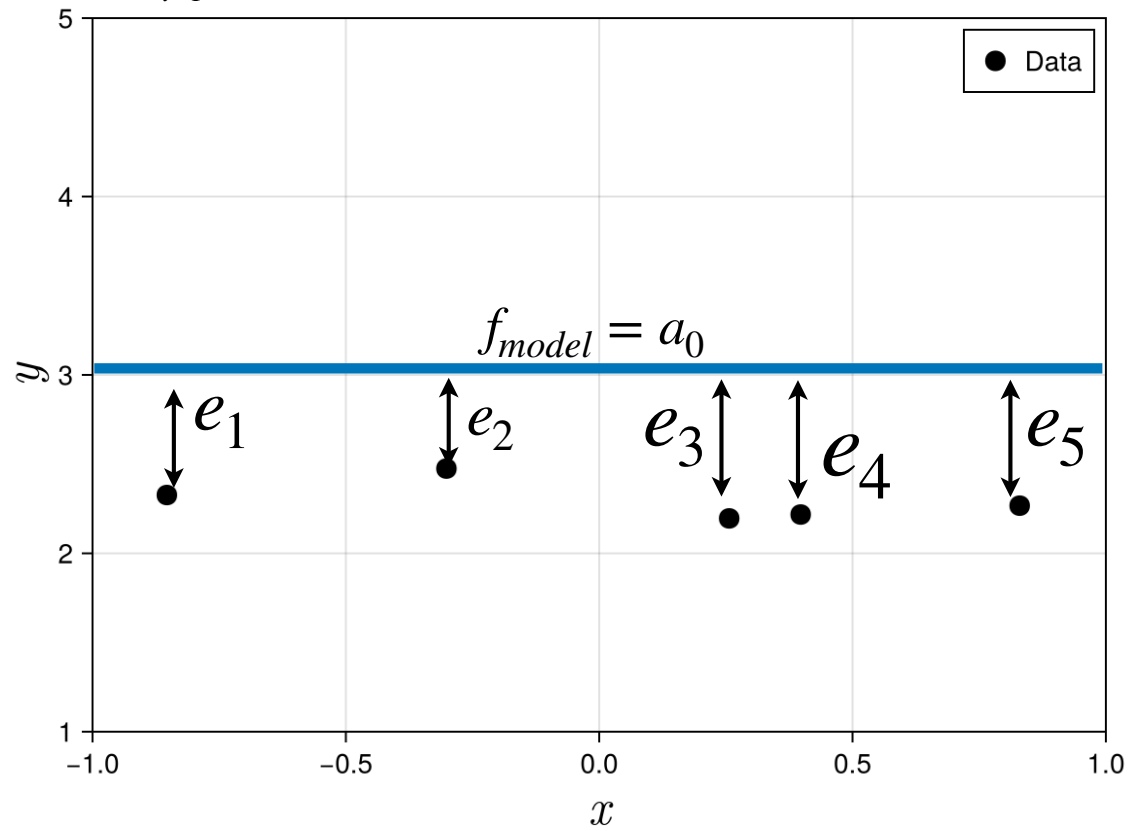
Define “mean” error to be

$$S = \frac{1}{2}(e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2)$$
$$= \sum_{i=1}^5 e_i^2$$



Define “mean” error to be

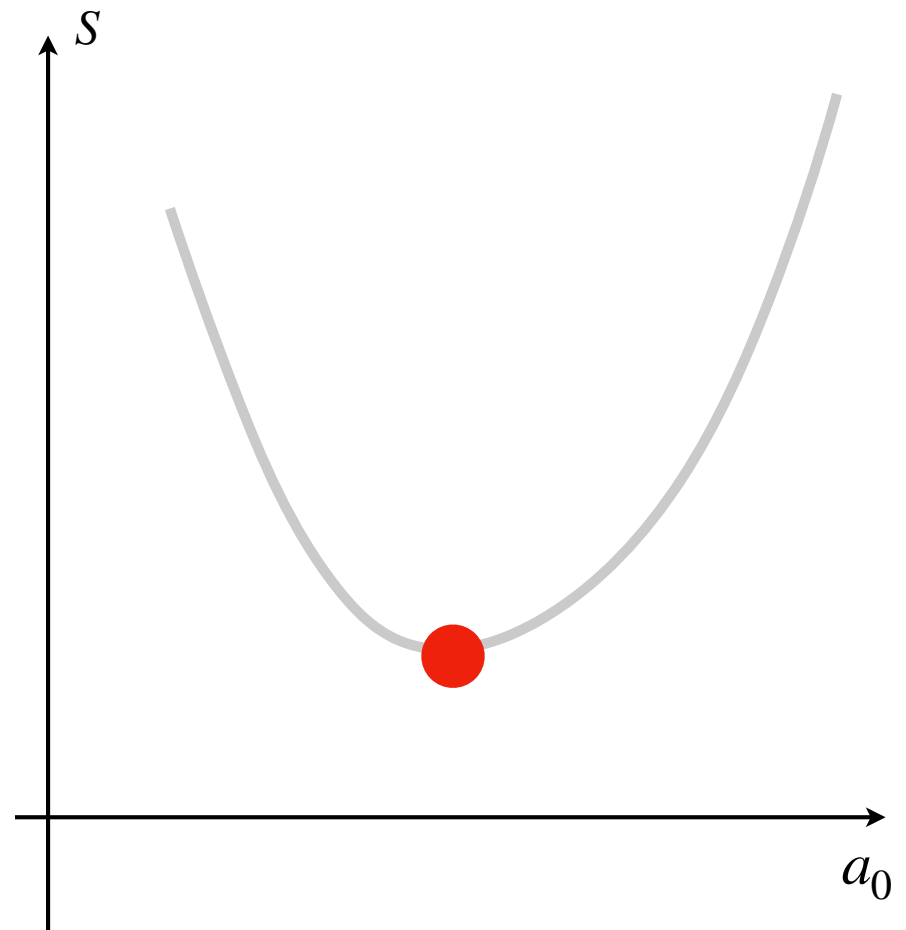
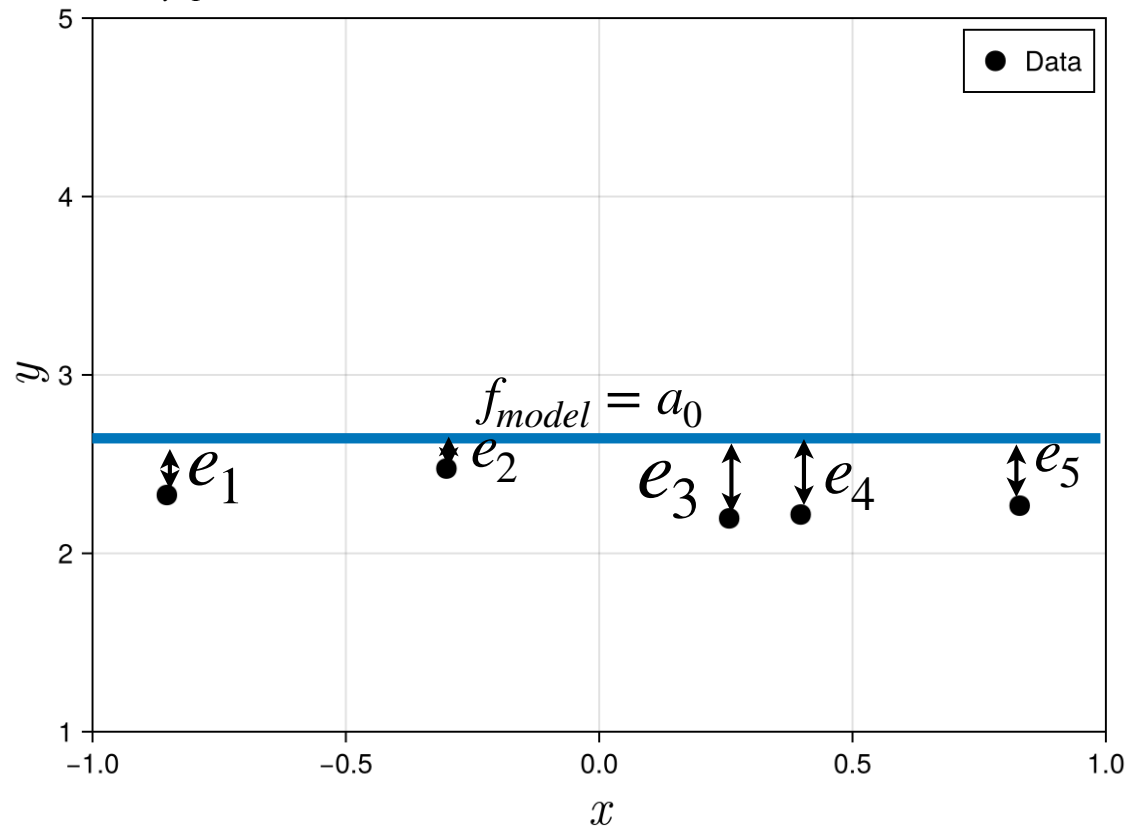
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Define “mean” error to be

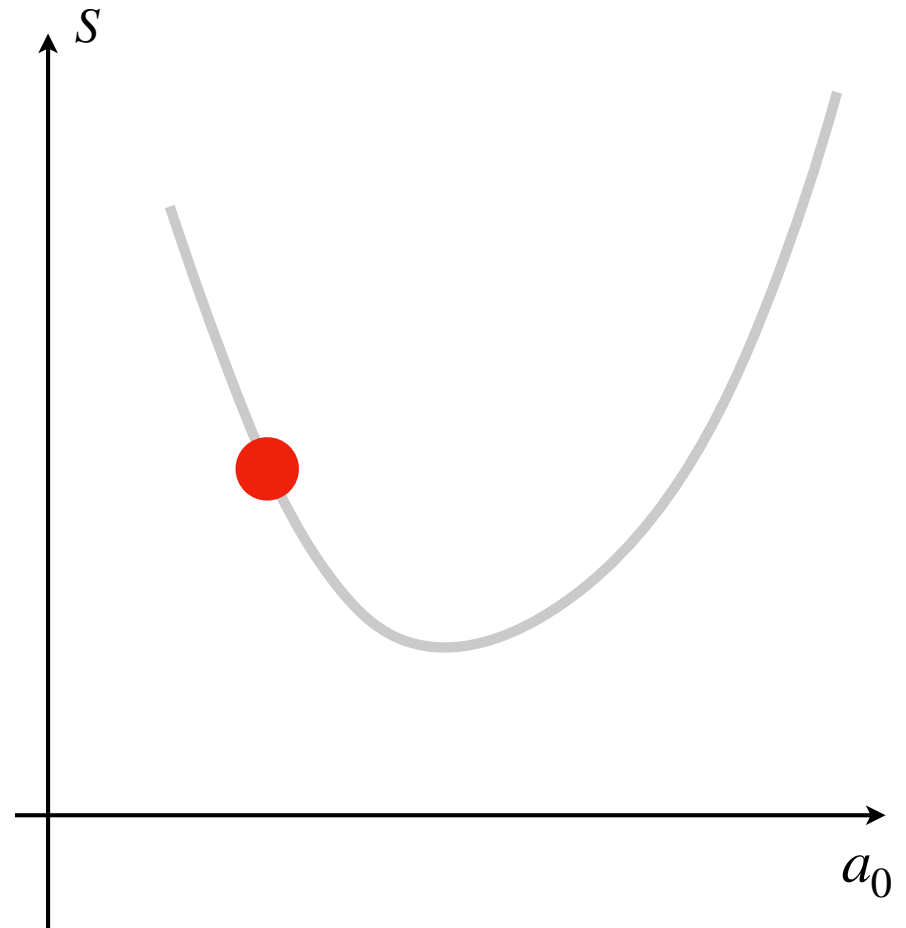
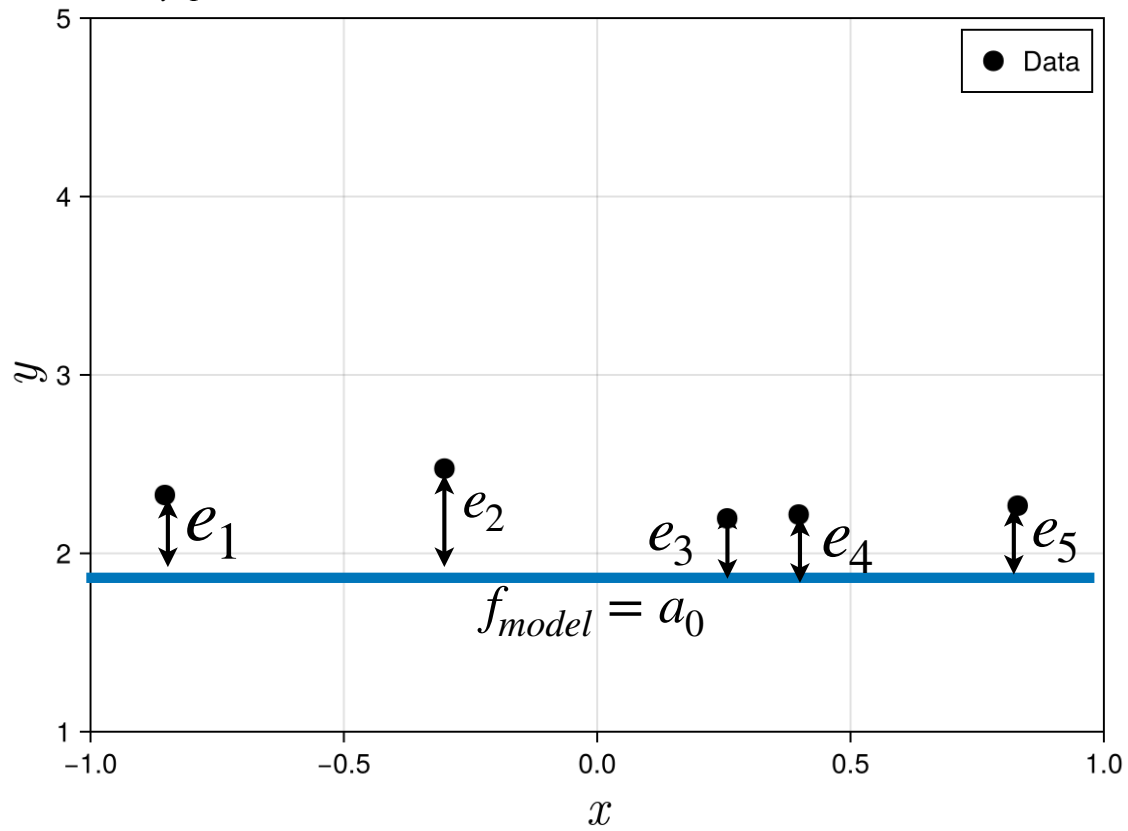
$$S = \frac{1}{2}(e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2)$$

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Define “mean” error to be

$$S = \frac{1}{2}(e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2)$$
$$= \sum_{i=1}^5 e_i^2$$



To find the minimum, we need to find the gradient of S with respect to a_0

$$S = \frac{1}{2}(e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2)$$

$$S = \frac{1}{2} ((y_1 - a_0)^2 + (y_2 - a_0)^2 + (y_3 - a_0)^2 + (y_4 - a_0)^2 + (y_5 - a_0)^2)$$

$$\frac{\partial S}{\partial a_0} = -((y_1 - a_0) + (y_2 - a_0) + (y_3 - a_0) + (y_4 - a_0) + (y_5 - a_0))$$

In general, if we have N data points, then

$$S = \sum_{i=1}^N e_i^2$$

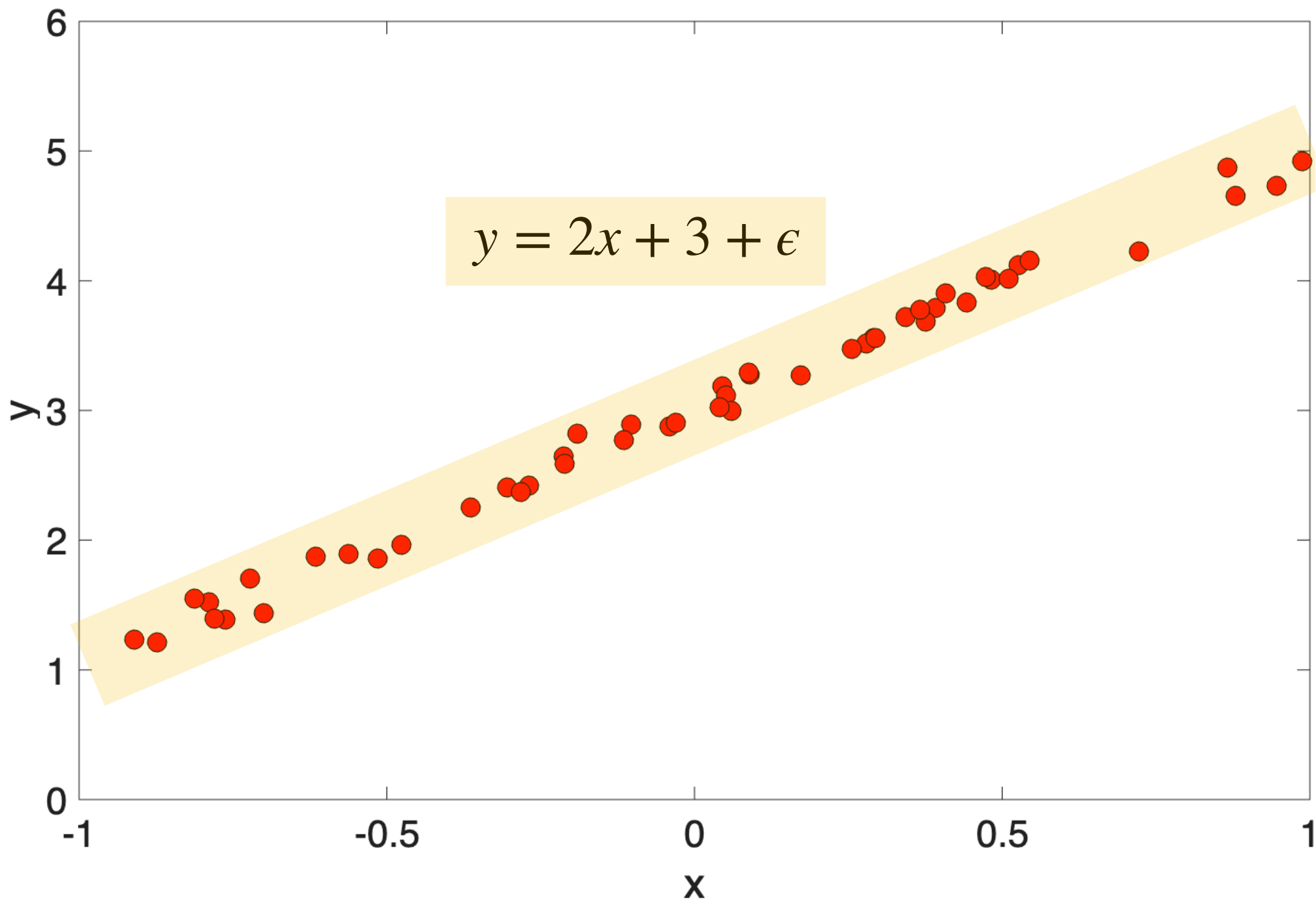
$$\frac{\partial S}{\partial a_0} = -\sum_{i=1}^N (y_i - a_0)$$

Let's look at a slightly more complex (fictitious) data set.

$$y = 2x + 3 + \epsilon$$

where $x \in [-1, 1]$. ϵ is some small amount of noise.

- For most problems, you do not know how the function looks like.
- But here, I am going to generate data from a known function and then I will find a model that fits the data.
- The advantage of having data from a known function is that I can tell if my model is correct and check if my process is right.
- For real problems, your data would come from your experiments, simulations and be stored in a spreadsheet.



We will define a model of the same form as the data.

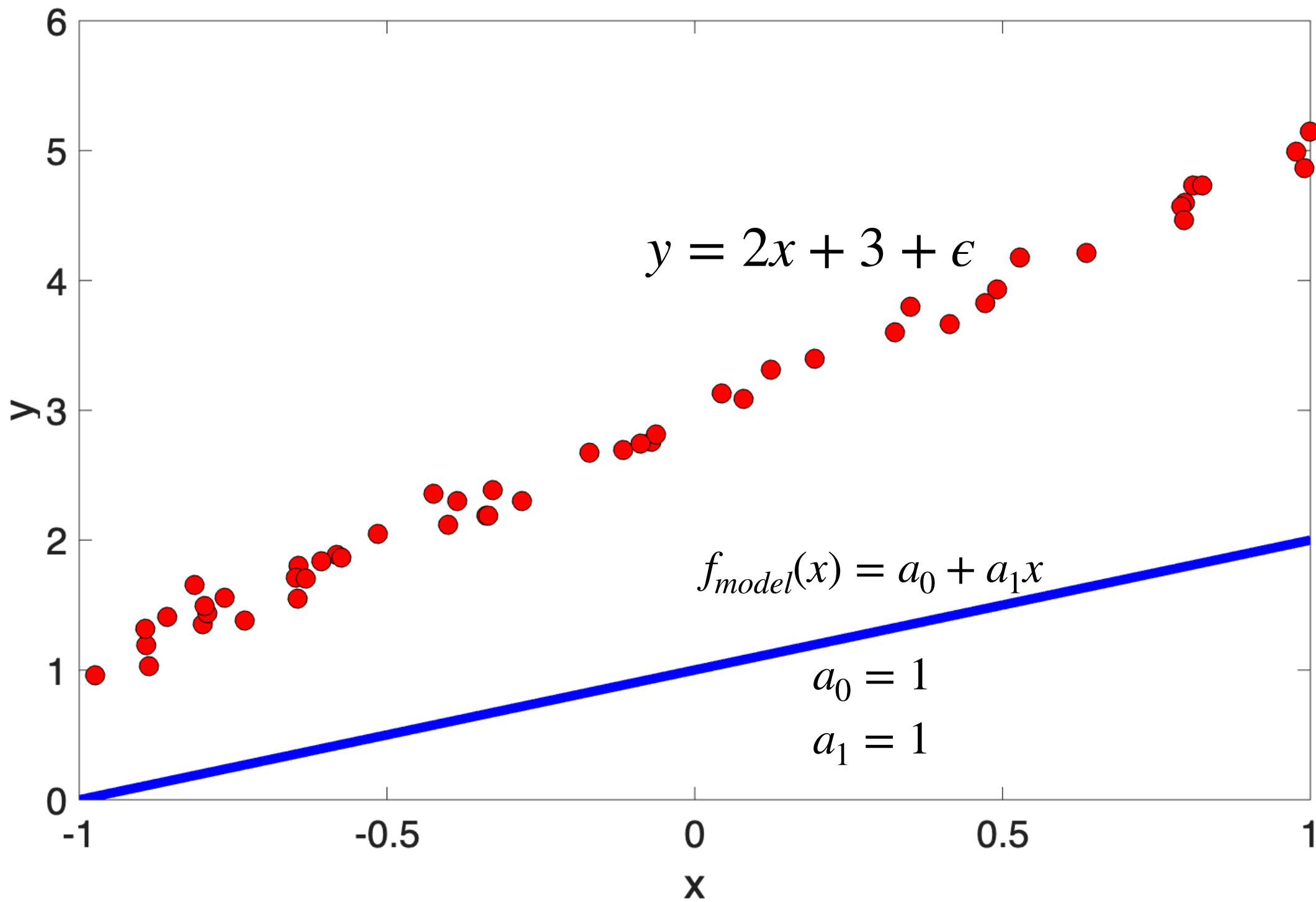
$$f_{model}(x) = a_0 + a_1x$$

Hence, our task now is to find a_0 and a_1 such that $f_{model}(x)$ closely approximates (x_i, y_i) .

Since we fictitiously generate the data, we know that f_{model} should fit the data best if $a_0 = 3$ and $a_1 = 2$

Note that in real world example, we usually do not know a_0 and a_1 .

We will now pretend we do not know the value of a_0 and a_1 and assume that they are both 1.0. Let's plot $f_{model}(x)$ and see how it looks like if we assume that $a_0 = a_1 = 1.0$.



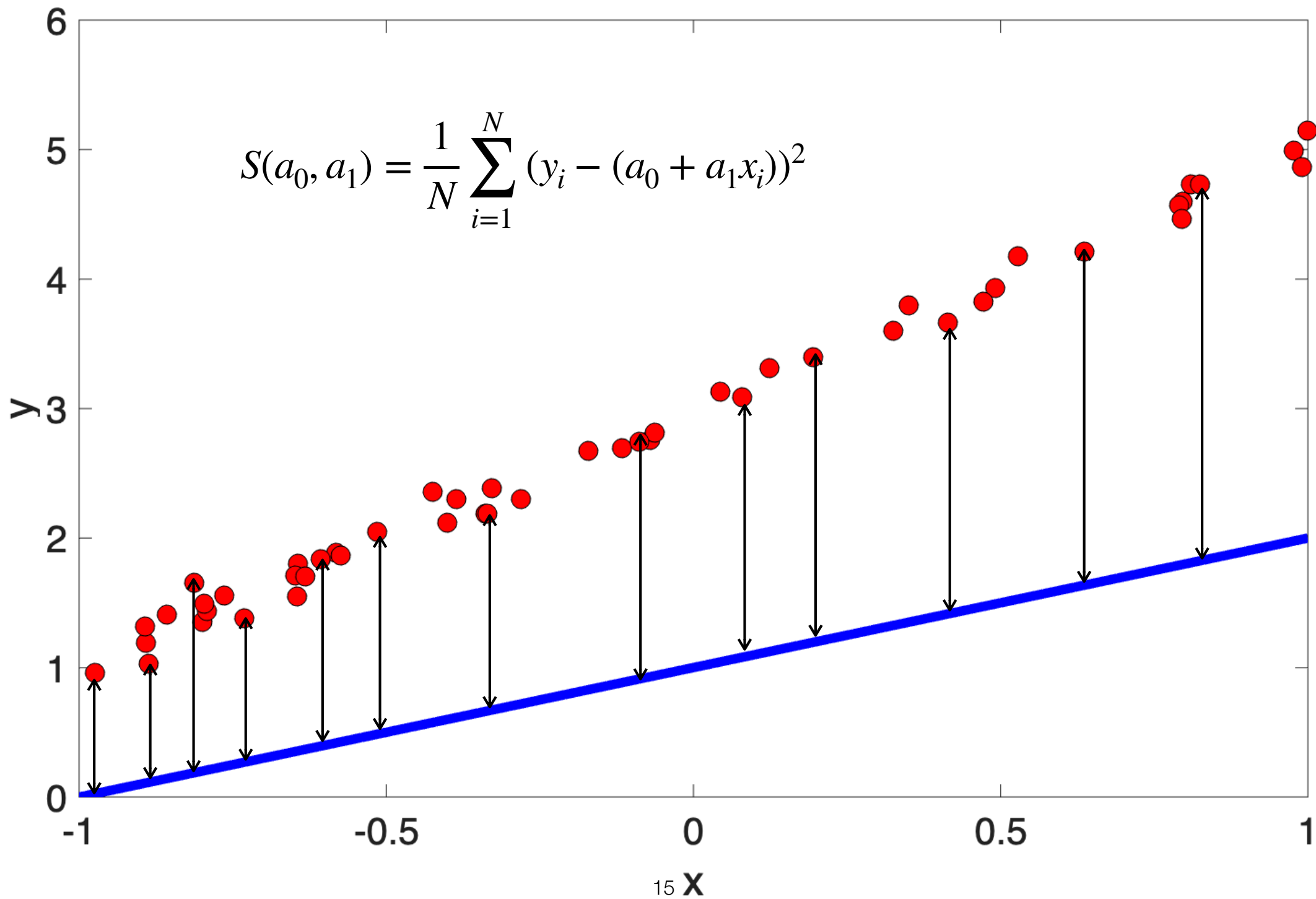
To find the optimum values, we will need to first define the error function

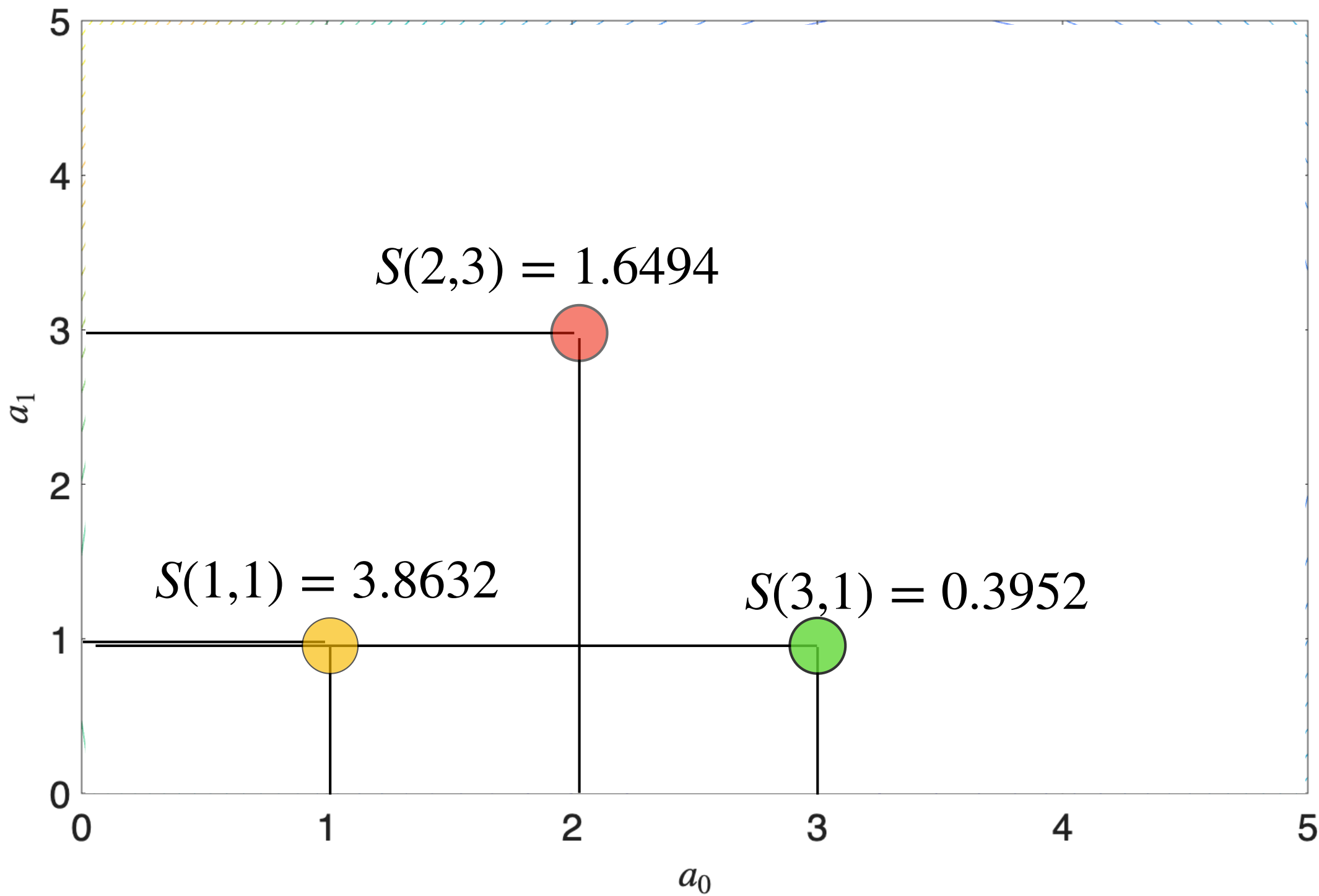
$$S(a_0, a_1) = \frac{1}{N} \sum_{i=1}^N (y_i - f_{model}(x_i))^2$$

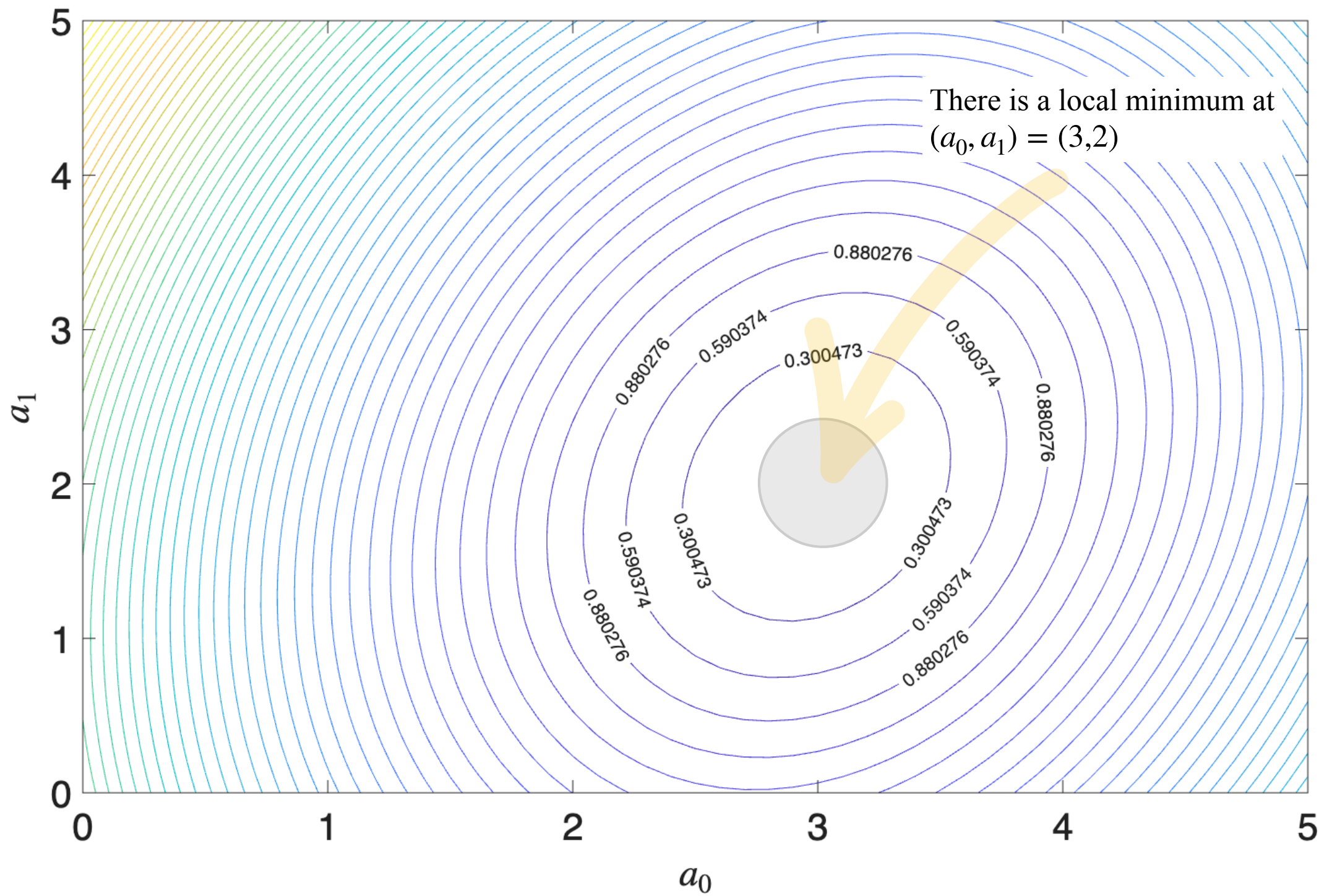
$$S(a_0, a_1) = \frac{1}{N} \sum_{i=1}^N (y_i - (a_0 + a_1 x_i))^2$$

We would then find values of a_0 and a_1 that minimises $S(a_0, a_1)$.

For this example, we will use standard gradient descent method.







Use standard gradient descent to find the minimum of $E(a_0, a_1)$

$$\frac{\partial S}{\partial a_0} = \sum_{i=1}^N 2(y_i - a_0 - a_1 x_i)(-1)$$

$$\frac{\partial S}{\partial a_1} = \sum_{i=1}^N 2(y_i - a_0 - a_1 x_i)(-x_i)$$

$$\begin{Bmatrix} a_0^{(k+1)} \\ a_1^{(k+1)} \end{Bmatrix} = \begin{Bmatrix} a_0^{(k)} \\ a_1^{(k)} \end{Bmatrix} + \alpha \begin{Bmatrix} \frac{\partial E}{\partial a_0}(a_0^{(k)}, a_1^{(k)}) \\ \frac{\partial E}{\partial a_1}(a_0^{(k)}, a_1^{(k)}) \end{Bmatrix}$$